Sant Dnyaneshwar Shikshan Sanstha's

ANNASAHEB DANGE COLLEGE OF ENGINEERING AND TECHNOLOGY, ASHTA

(An Empowered Autonomous Institute)

AVISHKAR 2026: PROJECT PRESENTATION AND REVIEW

Department of Computer Science and Engineering

TITLE:

Diagnosing Failure Geometry in Open-Weight LLMs: A Controlled Probe Suite for Long-Range Dependency Resolution

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Problem Statement:

The Problem

1. Aggregate scores are non-diagnostic
Benchmarks such as MMLU, HellaSwag and GSM8K reduce diverse model behaviour to a single score.
2. Failure behaviour remains hidden
A benchmark score does not show which inputs failed, where the failure occurred, or what form the failure took.
3. Architecture cannot be isolated directly
Independently trained models differ in training data, tokenizer and alignment, so behavioural differences cannot be attributed causally to architecture.
4. Controlled capability-level evaluation is needed
A diagnostic experiment should isolate one capability and deliberately vary one controlled factor.

Research Need

A reusable framework that studies failure behaviour under controlled conditions rather than only reporting overall accuracy.



Introduction & Proposed Solution :

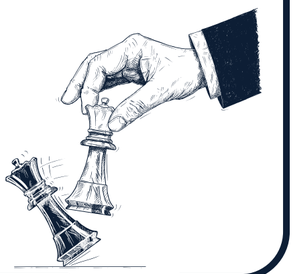
From “Which model scores higher?” to “Where and how does each model fail?”

Research Focus :

We study one capability:

Long-Range Dependency Resolution

A fact is introduced early in a long context, followed by distracting content, and the model is asked about that fact later.



Controlled Comparison Open Weight LLMs

Llama-2-7B

Full Attention

VS

Mistral-7B

Sliding-Window Attention (~4096-token window)



Proposed Solution

1. Controlled Probe Suite

Create ~500 probes with the fact-query distance varied across five bins.

- **Same probe → Same conditions → Different distance → Compare failures**

2. Reproducible Inference Harness

Run both models with llama.cpp, Q4_K_M, fixed seed and temperature = 0.

3. Failure-Level Diagnosis

Preserve exact outputs and classify failures using a structured, mutually exclusive taxonomy.

4. Comparative Failure Analysis

Do the two models fail on the same inputs or on different inputs as dependency distance increases?

Objectives :

01	Probe Suite Construction	Build approximately 500 controlled probes isolating dependency distance.
02	Reproducible Inference Harness	Deploy both models using identical inference conditions.
03	Failure Curve Recovery	Measure recall accuracy across the five distance bins.
04	Failure Taxonomy	Classify failures into mutually exclusive classes
05	Divergence Analysis	Measure overlap between model failure sets using the Jaccard index.
06	Open Artifact Release	Release the probe suite, inference harness and result explorer as reusable artifacts.

Literature Review & Research Gap :

Research Area	Key Work	Contribution to Our Study
Transformer Attention	Vaswani et al., 2017	Full self-attention foundation
Efficient Attention	Longformer, 2020	Local/sliding attention concept
Llama 2	Touvron et al., 2023	Open-weight full-attention model
Mistral 7B	Jiang et al., 2023	Sliding-window attention
Behavioural Evaluation	CheckList, 2020	Testing beyond aggregate accuracy
Long Context	Lost in the Middle	Position-dependent failures
Stress Testing	RULER / BABILong	Long-context capability evaluation

Research Gap

Existing studies provide:

Architecture studies
+
Benchmark evaluation
+
Behavioural testing
+
Long-context testing

- **But our study combines them into:**

Controlled dependency-distance probing + explicit failure taxonomy + cross-model failure-set comparison

Scope & Experimental Methodology :

✓ IN SCOPE

Two open-weight models with divergent attention architecture

Llama3.1-8B (full attention) vs Qwen3-8B (sliding-window)

One capability axis: long-range dependency resolution

Dependency distance binned into 5 levels; all else held constant

Text-only, English, inference-only evaluation

No training, no fine-tuning, no multimodal input

Four deliverables

Probe suite · Inference harness · Failure taxonomy · Result explorer

✗ OUT OF SCOPE

Causal attribution to architecture

Training data, tokenizer, and alignment are uncontrolled confounds — we report descriptive, not causal findings.

Training or fine-tuning

Compute-infeasible on our hardware; outside the research question entirely.

Multimodal or creativity evaluation

Different model class — not comparable with our text-only probe framework.

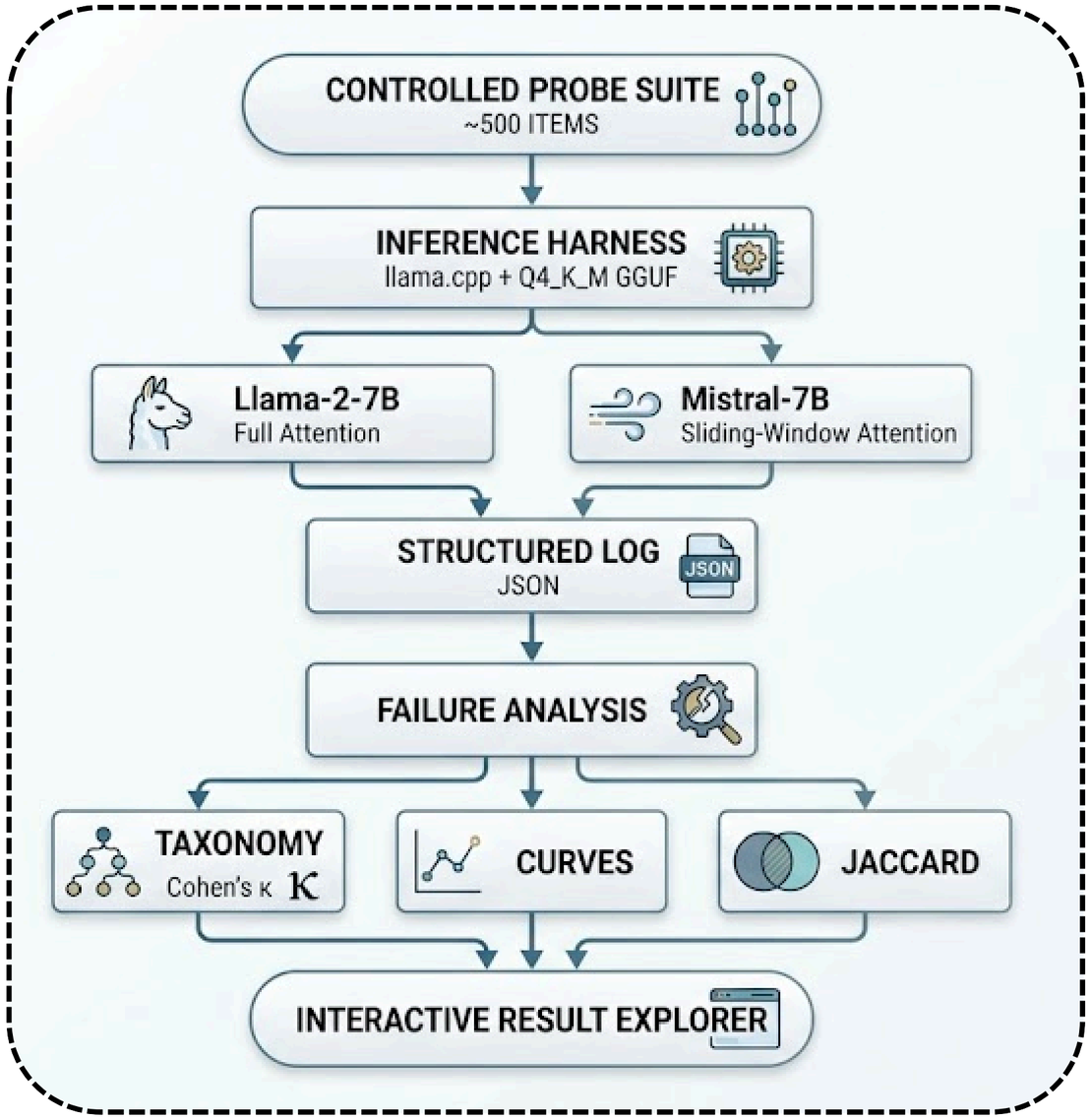
Leaderboard scoring

We report distributions and failure classes, not rankings or benchmarks.

Experimental Method

Probe Template
↓
Dependency-Distance Bins
↓
Probe Generation
↓
Human Solvability Check
↓
Reproducible Inference
↓
JSON Logging
↓
Failure Annotation
↓
Statistical / Comparative Analysis

System Architecture & Evaluation :



Evaluation Output:

- Recall per distance bin
→ How retention changes with distance

- Failure-class distribution
→ What type of failure occurs

- Cohen's κ
→ Agreement between annotators

- Jaccard index
→ Overlap between model failure sets

References :-



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[7] Liang et al., Holistic Evaluation of Language Models (HELM), 2022.
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[10] Kamradt, Needle In A Haystack: Pressure Testing LLMs, 2023.
[11] Liu et al., Lost in the Middle: How Language Models Use Long Contexts, TACL, 2024.
[12] Hsieh et al., RULER: What's the Real Context Size of Your Long-Context Language Models?, 2024.
[13] Kuratov et al., BABILong: Testing the Limits of LLMs with Long Context Reasoning-in-a-Haystack, 2024.

Conclusion & Expected Outcomes :

What We're Building :

A diagnostic instrument—not a leaderboard—that reveals where and how LLMs fail through controlled probes and failure analysis.

How It Works :

Inference-only

• Q4_K_M quantization • Owned hardware • No paid compute

Reproducible experiments with all prompts, outputs and analysis artifacts preserved.

Expected Outcomes :

- Open Probe Suite
- Failure Taxonomy + Cohen's κ
- Failure Curves + Jaccard Analysis
- Interactive Result Explorer
- Open GitHub Artifacts
- Research Paper / Report

Thank You...!

"We are not ranking models. We are building an instrument that reveals where and how they break."